PassFrame tuition

Number Sequences & Patterns

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 5n − 1

2. 51

3. A — 6n+1 (n=1 gives 7)

4. 3, 8

5. 26

6. A, C and E — common difference of 9

7. 3, 10, 17

8. 155, 140

9. C — 84

10. 0, 10

11. 5, 5.5

12. A, B, C and E — 13, 19, 25 and 37 are all terms (30 is not)

13. 109 − 9n (or −9n + 109)

14. B — 33

15. 40, 57

Standard

1. 3, 8, 15

2. 162, 486

3. C — 75

4. 135

5. 62.5

6. A, C and D — each term is ×3

7. 46

8. n^2; 144

9. A — $2n^2$

10. 189

11. A, B, C and E — 3, 8, 15 and 24 are all terms (20 is not)

12. 100, 50

13. 42

14. D — 162

15. 768

Standard / Hard

1. 7n + 5; 50th term = 355

2. B — n = 27 (4×27−7 = 101)

3. 1,260 seats

4. A, C and E — all match 8n − 5

5. 91

6. 515 − 15n; the 35th term is −10

7. C — 12

8. ≈78cm

9. n = 5; both terms equal 29

10. A, B and D — 1, 8 and 21 are terms (n=1,2,3)

11. 165 seats

12. $n^3 + 1$

13. B — the 25th term (100 − 4×25 = 0)

14. ≈£281

15. n = 20 (the 20th triangular number is 210)

Homework — Answers

Easy / Standard

1. 7n − 4

2. 85

3. A — 7n+2 (n=1 gives 9)

6. A, B, D and E — common difference of 11

7. 3, 12, 21

8. 234, 212

11. 7, 7.75

12. A, B, D and E — 11, 19, 27 and 35 are all terms (25 is not)

13. 163 − 13n

4. 1, 8

5. 24

9. C — 111

10. −2, 10

14. B — 43

15. 53

Standard

1. 4, 10, 18

2. 243, 729

3. C — 100

4. 108

5. 125

6. A, C and D — each term is ×4

7. 47

8. $(n+1)^2$; 169

9. A — $3n^2$

10. 320

11. A, B, C and E — 5, 12, 21 and 32 are all terms (30 is not)

12. 200, 100

13. 29

14. C — 135

15. 1,215

Standard / Advanced

1. 117 ($4\times30-3$)

2. B — 97 (10^2-3)

3. n = 7 (both terms equal 51)

4. A, B, D and E — 14, 23, 41 and 50 are all terms (30 is not)

5. n = 34 ($200-6\times34=-4$)

6. 1,620 seats

7. 121 sticks

8. A — ≈3.43m (10×0.7^3)

9. n^3+2

10. A, B, C and E — 6, 21, 46 and 126 are all terms (80 is not)

11. Row 11 ($60-4\times10=20$)

12. ≈£11,100 ($18{,}000\times0.85^3=11{,}054.25$)

13. C — 36

14. 126 edges

15. 83 (3→5→11→29→83)